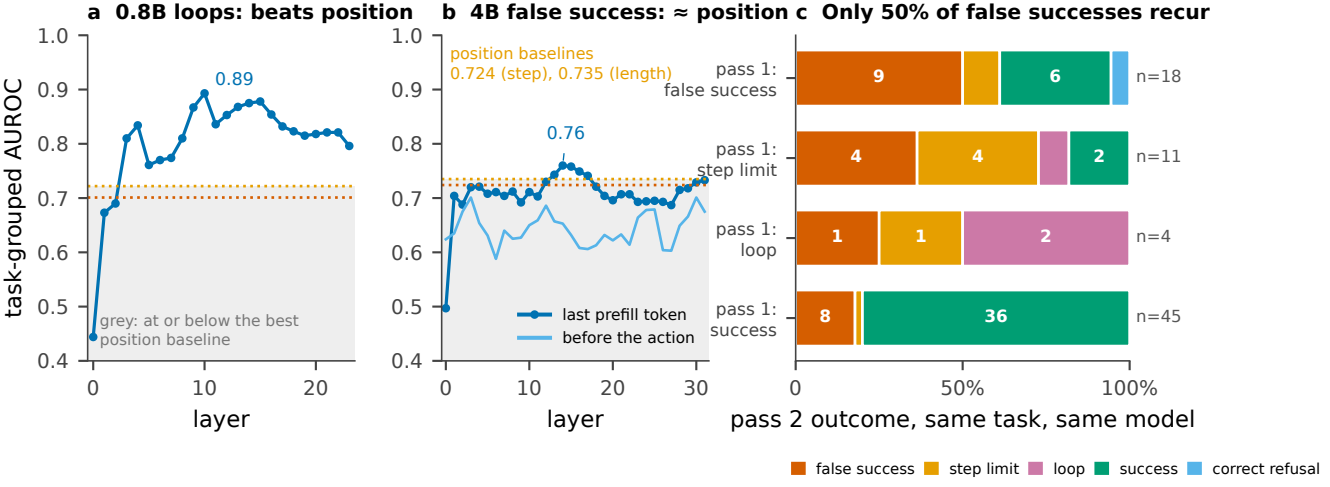
a 0.8B loops: beats position
b 4B false success: ≈ position c Only 50% of false successes recur
task-grouped AUROC
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.89
grey: at or below the best position baseline
0
10
20
layer
position baselines
0.724 (step), 0.735 (length)
0.76
last prefill token
before the action
0
10
20
30
layer
pass 1: false success
pass 1: step limit
pass 1: loop
pass 1: success
9
6
n=18
4
4
2
n=11
1
1
2
n=4
8
36
n=45
0
50%
100%
pass 2 outcome, same task, same model
false success
step limit
loop
success
correct refusal